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M168_Notes_6.4

6.4 Combinations and Permutations

Example
How many ways can 10 students be seated in a row of 10 chairs?

$10!$

Example
How many ways can 4 of 10 students be seated in a row of 4 chairs?

$10 \times 9 \times 8 \times 7 \Rightarrow \frac{10!}{6!} = \frac{10\times9\times8\times7\times\cancel{6}\times\cdots\times2\times1}{\cancel{6}\times\cancel{5}\times\cancel{4}\times\cancel{3}\times\cancel{2}\times\cancel{1}} = 10\times9\times8\times7 = \frac{10!}{(10-4)!}$

Permutations: If we have a finite set of n elements and we want to place r of them in an arrangement, we say the number of permutations of n things arranged r at a time is $P(n, r)$.

$P(n,r)$, ${}_nP_r$, $\boxed{{}_nP_r = \frac{n!}{(n-r)!}}$

Example
How many ways can gold, silver and bronze medals be awarded in a race of 12 people?

$P(12,3) = \frac{12!}{9!} =$

Example
How many ways can a group of 4 students be chosen from 10 students?

$\frac{10!}{(10-4)!\times 4!} = C(10, 4)$

Combinations: If we have a finite set of n elements and we want to take r of them in an group, we say the number of combinations of n things grouped r at a time is $C(n, r)$.

$C(n,r) = \frac{n!}{(n-r)!\times r!}$

Example
How many ways can a hand of 6 clubs be chosen from a standard deck?

Example
From a group of 12 people, how many ways can a committee of 4 be formed if one person is the chair of the committee?

$C(12,4) = \frac{12!}{8! \times 4!}$

Chair 12 $\times C(11,3) =$

Example
A bag contains 6 blue, 1 green and 3 pink jelly beans. You choose 3 at random. How many samples are possible in which

a) the jelly beans are all blue? $\rightarrow C(6,3) = \frac{6!}{3! \times 3!} = \frac{6 \times 5 \times 4}{3 \times 2} = 20$

b) the jelly beans are all green? $\rightarrow 0$

c) the jelly beans are all pink? $\rightarrow 1 = C(3,3) = \frac{3!}{0! \times 3!} = 1$

d) there are 2 blue and 1 pink? $C(6,2) \cdot C(3,1) = 45$

e) How many ways to choose 3 jelly beans? $C(10,3) = 120$

f) How many ways to choose no blue? $C(4,3) = 4$

g) How many ways to choose at least one blue?

$C(6,1) \times C(4,2) + C(6,2)\, C(4,1) + C(6,3) = 116$

Example
A class of 12 students will divide into 3 teams of 4. How many ways can this be done?

$1, \ldots, 12$

$C(12,4) \times C(8,4) \times C(4,4) =$ $1 \ldots 4,\ 5 \ldots 8,\ 9 \ldots 12.$

Example
A school is putting together a committee. The committee will have a chair and an assistant chair chosen from a group of 10 teachers, two parents chosen from a group of 15 parents and two students chosen from a group of 20 students. How many different committees are possible?

Example
You are dealt a hand of four cards from a well-shuffled standard deck of 52 cards.
(a) How many ways can you be dealt at least 3 spades?

(b) How many ways can you be dealt exactly two diamonds or exactly two clubs?

Example
How many different "words" can be made from the letters in MISSISSIPPI?

$$\boxed{\frac{11!}{4! \times 4! \times 2!}}$$

Example
Seven children stop at a restaurant where they have a choice of a cheeseburger, a hot dog, pizza or a burrito. How many different purchases are possible?